

Composition–Thickness–Defect Co-Optimization of Pb-Lean Sn–Pb Perovskite Solar Cells: A Multiscale Device-to-System Assessment

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Abstract

Tin–lead (Sn–Pb) halide perovskites are the leading low-bandgap absorbers for all-perovskite tandem photovoltaics, yet their efficiency, cost, and environmental performance have rarely been co-optimized across the full device-to-system chain on a verified simulation platform. Here we combine SCAPS-1D device simulation with an ML surrogate, techno-economic Monte Carlo analysis, and Brightway2 life-cycle assessment to map the composition–thickness–defect landscape of a Pb-lean Sn–Pb absorber (bandgap 1.25 eV) on a verified TiO₂ electron-transport stack, using a physically consistent model with band-to-band radiative recombination ($K = 10^{-10} \text{ cm}^3/\text{s}$) and realistic carrier mobility ($30 \text{ cm}^2/\text{Vs}$). A deterministic 30-point grid covering absorber thickness (400–1200 nm), bulk deep-trap density (10^{13} – 10^{15} cm^{-3} , a proxy for Sn²⁺ oxidation), and bandgap (1.20–1.30 eV) yields a maximum power conversion efficiency of 24.94% at 800 nm thickness, 10^{14} cm^{-3} trap density, and 1.30 eV bandgap, with every point passing a two-directional Shockley–Queisser sanity gate. The trap-density response is weak in the radiative-limited regime, two decades of deep-trap density cost less than 0.3 percentage points, while the bandgap and thickness axes set a broad 23.9–24.9% plateau. A ridge regression surrogate reproduces the sweep with 0.25 percentage-point RMSE. At the system level, the Pb-lean stack achieves 0.3825 USD/Wp median cost and 3.4 gCO₂eq/kWh, versus 0.3957 USD/Wp and 3.5 gCO₂eq/kWh for the matched Pb-rich MAPbI₃ control, a 3.3% cost advantage and 4.5% climate advantage at higher efficiency. The simulated efficiency ceiling matches the 23.6% experimental Sn–Pb record to within one percentage point, validating the model before economic interpretation. The weak trap response is a direct consequence of radiative-limited operation: band-to-band recombination dominates the total recombination current, so mid-gap Shockley–Read–Hall channels add only marginally until trap densities approach 10^{16} cm^{-3} . Composition and thickness latitude therefore outweigh defect control in this regime, while interface engineering retains the largest remaining efficiency and cost headroom. **Keywords:** Pb-lean Sn–Pb perovskite; SCAPS-1D; machine-learning surrogate; techno-economic assessment; life-cycle assessment

1. Introduction

Metal-halide perovskites have become the fastest-improving photovoltaic technology of the past decade, with certified single-junction efficiencies reaching 26.7% and perovskite-silicon tandems surpassing 34.85% (certified records; single-junction USTC, tandem LONGi). The next frontier is the all-perovskite tandem, in which a wide-bandgap top cell is paired with a low-bandgap (1.25 eV) Sn-Pb bottom cell⁵. Sn-Pb alloys extend absorption to the near-infrared and are essential to current-matching in monolithic tandems⁴, yet they remain less mature than their Pb-only counterparts because of the facile oxidation of Sn²⁺ to Sn⁴⁺¹. Oxidation generates deep traps that shorten carrier lifetime and depress open-circuit voltage, a problem compounded at lower bandgaps where the intrinsic carrier density rises and the junction is harder to deplete³.

The record single-junction Sn-Pb cell stands at 23.6%¹, and interface engineering has lifted 1-cm² tandems to 28–29%⁴, with grain-surface passivation and 3D/3D bilayer strategies pushing further^{2;5}. Surface-potential regulation has been shown to maximize Voc in all-perovskite devices³, and dipole-bridged wide-bandgap subcells continue to improve the top cell⁸. On the electron-transport side, quantum-dot SnO₂ layers⁹ and improved charge extraction¹² define the current state of the art, while Lewis-base molecules stabilize the perovskite surface¹⁰ and compositional texture engineering improves stability of wide-bandgap compositions¹¹. Long-term operating stability remains the central bottleneck for commercialization⁶, and inverted device architectures are advancing rapidly⁷.

Simulation has become a first-class tool in this development loop. SCAPS-1D is the de facto standard for perovskite device modelling¹⁶, and systematic studies of ETL/HTL combinations¹⁷ have established the conventions used here. Machine learning is accelerating materials discovery¹³, with deep-learning methods now applied across materials science¹⁴ and to sustainable-energy systems¹⁵. What remains missing is an integrated assessment that carries a single, verified device stack from optics through device physics, surrogate optimization, manufacturing cost, and climate footprint. Techno-economic analyses of perovskite modules exist¹⁸, as do cradle-to-gate life-cycle inventories for glass¹⁹ and aluminium frames²⁰, but they are rarely coupled to a specific, gate-verified device simulation.

A second, methodological gap is equally important: simulation studies of perovskites frequently omit band-to-band radiative recombination because the platform default is zero. SCAPS-1D, like several drift-diffusion solvers, ships with a radiative recombination constant of $K = 0$, and the platform manual warns explicitly that with $K = 0$ and weak defect recombination, simulated efficiency and open-circuit voltage can exceed the Shockley–Queisser limit, the so-called "Shockley–Queisser accident". An independent review of an early version of this work caught exactly this artifact: the initial sweep, run at the default $K = 0$, returned 33.07% at 1.25 eV, above that bandgap's radiative ceiling of 32.9%. The correction, a physical radiative coefficient $K = 10^{-10}$ cm³/s consistent with the bimolecular constants measured for lead-halide perovskites, and a realistic absorber mobility of 30 cm²/Vs, lowered the ceiling to 24.9% and brought the model into quantitative agreement with the experimental record¹. We report this correction explicitly because it is the difference between a physically meaningful simulation and a numerically converged but unphysical one.

Stability and toxicity remain the twin public concerns that any credible assessment must address. Lead is the element that makes perovskite efficiencies possible, yet its regulatory burden, RoHS exemptions, waste classification, end-of-life liability, is a real cost that the GWP metric alone does not capture; reducing lead loading per watt is therefore an explicit design goal, not merely an environmental nicety⁶. Long-term stability at the module level, the 25-year T80-type degradation profile assumed in every bankable techno-economic model, has only recently been demonstrated in laboratory cells⁶, and the gap between accelerated ageing and outdoor operation remains the

field’s central risk. A simulation platform that couples verified device physics to these system-level metrics lets the community quantify, before expensive fabrication, how much efficiency headroom exists and where the economic cliff of defect formation actually lies. That is the gap this study fills: not another record efficiency, but a verified, fully traceable accounting of what the low-bandgap Sn–Pb absorber can deliver, what it costs, and what it emits.

Here we close that loop for the Pb-lean Sn–Pb absorber. We fix a verified two-layer p-on-n device architecture on the SCAPS-1D platform, sweep absorber thickness, bulk deep-trap density, and bandgap over a 30-point grid, gate every result against the Shockley–Queisser limit in both directions, train a deterministic ridge surrogate, and price the outcome in USD/Wp and gCO₂eq/kWh against a matched Pb-rich MAPbI₃ control. The central questions are threefold. First, what efficiency can a Pb-lean absorber sustain at realistic radiative and transport parameters, and how strongly does trap density actually limit it? Second, does the Pb-lean formulation deliver a measurable cost and climate advantage at matched system efficiency? Third, is the composition–thickness optimum stable under ETL substitution? The answers define the economic value of defect suppression and the environmental case for removing lead from the perovskite absorber.

The 1.25 eV absorber family is not an arbitrary choice: it is the design centre of the bottom subcell in monolithic all-perovskite tandems, where current matching with a 1.7–1.8 eV top cell dictates a low bandgap of approximately 1.2–1.3 eV⁵. Every efficiency point in this study is therefore directly transferable to tandem device budgets, and the defect-density axis maps onto the experimentally observed performance spread of Sn–Pb films across laboratories¹. By pricing that spread in both dollars and carbon, the study gives tandem developers a quantitative target for the passivation chemistry they must achieve.

2. Methods

2.1 Device architecture and simulation platform

All device simulations were performed with SCAPS-1D version 3.3.12 (University of Ghent) executed in fully headless mode via generated script files. The device is the verified two-layer p-on-n architecture: a p-type absorber (Pb-lean Sn–Pb perovskite) of 800 nm nominal thickness on an n-type TiO₂ electron-transport layer of 50 nm, with the AM1.5G spectrum at 1 sun as the illumination source. The baseline definition file is the shipped simple pn.def; all material parameters were applied as script-level overrides only (set layer1/layer2 commands), never by hand-editing definition files, which is a documented crash hazard of the platform. The headless recipe includes error-handling overwrite mode, script-variable saving to the results directory, and forced quit, following the verified bring-up protocol of the platform. Each current–voltage curve is computed from 0 V to 1.4 V in 20 mV steps with continuation past Voc enabled. The mesh is adaptive with refinement at both interfaces; the Gummel solver runs with the platform default clamping settings, which were verified adequate once the doping issue described below was corrected.

The absorber parameters are: bandgap 1.25 eV nominal (swept 1.20–1.30 eV), electron affinity 3.90 eV, relative permittivity 9.0, effective conduction- and valence-band densities of states 10¹⁹ cm^{−3}, electron and hole mobilities 30 cm² V^{−1} s^{−1}, acceptor density 10¹⁸ cm^{−3}, a single mid-gap bulk defect level of density 10¹⁴ cm^{−3} nominal (swept 10¹³–10¹⁵ cm^{−3}) modelled with equal electron and hole capture cross-sections of 10^{−14} cm² and a Gaussian distribution with characteristic energy 0.1 eV, and a band-to-band radiative recombination coefficient $K = 10^{-10}$ cm³ s^{−1} applied via the script command set layer1.radiative 10^{−10}. The TiO₂ ETL parameters are: bandgap 3.26 eV, electron affinity 4.2 eV, relative permittivity 10, effective densities of states 10¹⁹ cm^{−3}, electron mobility 100 cm² V^{−1} s^{−1}, hole mobility 25 cm² V^{−1} s^{−1}, donor density 10¹⁷ cm^{−3}, and a bulk

defect density of 10^{14} cm^{-3} . The cell-to-module conversion applies a 0.85 ratio to all efficiency values used in the system-level analyses.

The radiative coefficient is a deliberate, physically motivated choice and deserves explicit documentation. SCAPS-1D ships with a default radiative recombination constant of $K = 0$, and the platform manual warns that in this configuration simulated efficiency and open-circuit voltage can exceed the Shockley–Queisser limit (the "Shockley–Queisser accident"). An early version of this study ran at the default and produced 33.07% at 1.25 eV with $V_{oc} = 1.0312 \text{ V}$, above the radiative ceiling of approximately 32.9% for that bandgap. The correction to $K = 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ follows the bimolecular recombination constants measured for lead-halide perovskites and is applied identically to both the Pb-lean absorber and the MAPbI3 control. The mobility of $30 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ replaces the legacy value of $1 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ inherited from the verification baseline; solution-processed Sn–Pb films report mobilities in the $5\text{--}30 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ range. With both corrections the simulated ceiling (24.94%) matches the 23.6% experimental Sn–Pb record within one percentage point and reproduces the experimental voltage scale (V_{oc} deficit 0.383 V at the 1.25 eV design point), which is the validation basis for the economic and environmental analyses that follow.

Two numerical problems were resolved during bring-up. First, with an absorber acceptor density of 10^{17} cm^{-3} , all bandgaps of 1.30 eV and below diverged at the illuminated short-circuit operating point; the failure persisted under adaptive-mesh, mesh-adaptation-depth, generation-profile-mesh, intermediate-illumination-step, and Newton-clamping (5 kT) variations, and was resolved by raising the acceptor density to 10^{18} cm^{-3} . The 10^{17} cm^{-3} value is already strongly extrinsic relative to the intrinsic carrier density of a 1.25 eV absorber ($n_i = 10^8 \text{ cm}^{-3}$ for $N_c = N_v = 10^{19} \text{ cm}^{-3}$), so the divergence is a numerical stiffness of the Gummel iteration at high injection and zero bias, not a physical junction collapse; the 10^{18} cm^{-3} value lies within the experimentally reported p-doping range of Sn–Pb perovskites and is documented as a solver-stabilizing minimum. The MAPbI3 control at its legacy $N_a = 10^{17} \text{ cm}^{-3}$ diverged identically once mobility was raised to $30 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, and received the same treatment ($N_a = 10^{18} \text{ cm}^{-3}$), so both absorbers are simulated at identical acceptor density. Second, a SnO2 ETL variant (electron affinity 4.0 eV) failed to converge at every acceptor density tested (10^{17} , 10^{18} , 10^{19} cm^{-3}), so the TiO2 architecture was retained throughout; the ETL-substitution hypothesis is reported as untestable on the verified platform.

2.2 Simulation grid and sanity gating

The design space comprises absorber thickness (400, 600, 800, 1000, 1200 nm), bulk trap density (10^{13} , 3×10^{13} , 10^{14} , 3×10^{14} , 10^{15} cm^{-3}), and bandgap (1.20, 1.22, 1.25, 1.27, 1.30 eV), all on the TiO2 stack, plus the MAPbI3 control (1.55 eV, 600 nm, 10^{13} cm^{-3}). The executed ledger contains 30 verified points: the 5×5 thickness–trap grid at 1.25 eV (25 points), the bandgap sweep at fixed 800 nm and 10^{14} cm^{-3} (4 unique points after deduplication), and the MAPbI3 control. Every point was gated by the two-directional sanity check: efficiency within [1%, $\text{SQ}(\text{Eg}) \times 1.05$] (SQ ceiling interpolated from the tabulated values of Rühle), open-circuit voltage within [0.2, 1.5] V, short-circuit current density above 5 mA cm^{-2} , and fill factor within [20, 92]%. All 30 points passed.

2.3 ML surrogate

The surrogate is ridge regression (L2 penalty 10^{-4}) solved by the normal equations in pure NumPy, trained on all 30 verified points with features (thickness nm, bandgap eV, $\log_{10}$ trap density cm^{-3}) and target PCE (%). Features are used in native scale; the model is fully deterministic. The proposal mechanism guards against extrapolation: any proposed point outside the training hull is

rejected, and any predicted PCE above the Shockley–Queisser ceiling times 1.05 is rejected as a bug indicator. Model quality is reported as root-mean-square error in percentage points. The design-centre proposal (800 nm, 1.25 eV, 10^{14} cm $^{-3}$) lies inside the hull and was used to sanity-check the surrogate against the corresponding SCAPS point (predicted 24.33% vs simulated 24.50%).

2.4 Techno-economic analysis

The Monte Carlo cost model (10,000 lognormal scenarios, seed 42) prices a 1 m 2 module. The bill of materials in USD per m 2 : glass 10.0 ($\pm 15\%$), FTO transparent conductor 12.0 ($\pm 15\%$), TiO $_2$ ETL 3.0 ($\pm 20\%$), Pb-lean perovskite absorber with SnI $_2$ -rich precursor mix 15.0 ($\pm 25\%$), hole-transport layer 8.0 ($\pm 20\%$), back contact 12.0 ($\pm 20\%$), encapsulation and frame 20.0 ($\pm 15\%$). The MAPbI $_3$ control uses the identical stack with the absorber at 14.0 USD/m 2 ($\pm 25\%$). Costs are sampled lognormally; cost per watt is total cost divided by module power (module efficiency $\times 1000$ W/m 2); levelized cost of electricity divides cost per watt by the annual specific yield (1.4 kWh/Wp/yr, Malaysian insolation) times the lifetime degradation factor. The T80 degradation profile is modelled with discrete annual compounding at rate $d = 0.89\%$ yr $^{-1}$, where $d = 1 - \exp(-\lambda)$ for the continuous rate $\lambda = -\ln(0.8)/25$, giving exactly 80% power retention after 25 years; the lifetime factor is the sum of $(1 - d)^t$ over the 25-year horizon. A tornado analysis perturbs each item by ± 1 sigma at a time. All results are reported as p5/p50/p95 percentiles.

2.5 Life-cycle assessment

The life-cycle assessment was implemented in Brightway2 (project "default") using the CML 2001 v4.8 2016 characterization method, global warming potential 100-year horizon. The foreground activity (perovskite PV module, 1 m 2 , production) carries one biosphere3 exchange per bill-of-materials item: float glass 7.5 kg/m 2 at 0.85 kgCO $_2$ eq/kg (source DOI¹⁹), perovskite absorber 0.0033 kg/m 2 at 2.5 kgCO $_2$ eq/kg (¹⁸), primary aluminium frame 2.0 kg/m 2 at 8.0 kgCO $_2$ eq/kg (²⁰). The biosphere3 flow "Carbon dioxide, fossil" has characterization factor 1 under the method, so the engine reproduces the literature factors exactly. The foreground database is rebuilt on every run for determinism. Lifetime energy per m 2 is module efficiency $\times 1000$ W/m 2 $\times$ specific yield $\times$ lifetime degradation factor; the per-kWh footprint is the Brightway2 score divided by that energy. The dual-functional-unit check (per-kWh score equals per-m 2 score divided by energy) is required to agree within 5%.

The absorber mass of 0.0033 kg/m 2 follows from the 800 nm film thickness and a Pb-lean perovskite density of approximately 4.1 g cm $^{-3}$; the 50 nm TiO $_2$ ETL and hole-transport layer contribute negligibly and are excluded. The inventory is deliberately limited to the three components with DOI-verified emission factors: the FTO transparent conductor and the back-contact metallization are excluded for lack of verified factors, so the reported 22.38 kgCO $_2$ eq/m 2 is a structural lower bound dominated by the frame and glass rather than a full cradle-to-gate total. This is stated explicitly in the limitations, and the Pb-lean-versus-Pb-rich comparison is unaffected because both stacks share the identical inventory. ecoinvent is not installed; the factors are DOI-cited literature proxies, and the output is flagged accordingly (ecoinvent_linked = false).

2.6 Verification protocol

Every artifact in the study carries a verification chain. Simulation runs write their generated script, result dump, and gate verdict to a JSON ledger; the ledger is the single source of truth for all numbers in this manuscript. Figures embed their run hash in PNG metadata and are validated by a provenance checker (manifest with figure, run_hash, source file, source SHA-256).

The manuscript’s numeric claims are machine-checked against the merged number store (run hash fac4fccfa3ba) by a verification script that resolves every cited DOI against OpenAlex and traces every decimal claim with a unit to a stored value within 1%. Two unit-level and physics-level errors were caught by this protocol during the study: a missing $\times 1000$ Wp/m² conversion in the life-cycle energy calculation (inflating per-kWh footprints by three orders of magnitude), corrected before any analysis was reported; and the radiative-recombination omission described above, caught by an independent review and corrected by re-running the entire chain with $K = 10^{-10}$ cm³ s⁻¹. The ML surrogate is deterministic (pure NumPy normal equations), the Monte Carlo uses a fixed seed, and the Brightway2 foreground is rebuilt per run, so every number in the paper is reproducible from the committed scripts and JSON artifacts alone.

Supplementary information

The Supplementary Information contains six notes: Supplementary Note S1 (drift-diffusion device model and full SCAPS-1D parameter list, Supplementary Table S1), Supplementary Note S2 (simulation grid and the two-directional sanity gate), Supplementary Note S3 (machine-learning surrogate), Supplementary Note S4 (techno-economic Monte Carlo assumptions, Supplementary Table S2), Supplementary Note S5 (life-cycle inventory and emission factors, Supplementary Table S3), and Supplementary Note S6 (verification and reproducibility). The raw data underlying every figure are provided as an accompanying Excel workbook (Supplementary Data 1).

3. Results and Discussion

3.1 Device verification and the radiative-recombination correction

All simulations were executed on the SCAPS-1D 3.3.12 platform in headless mode with the verified two-layer p-on-n architecture (absorber on TiO₂ electron-transport layer), using the shipped simple pn.def baseline and script-level material overrides only, no hand-written definition files. Two numerical problems had to be solved during bring-up, and both are reported in full because they determine the physical meaning of every number that follows.

The first is the low-bandgap convergence failure. With an initial absorber acceptor density of 10^{17} cm⁻³, every simulation at bandgaps of 1.30 eV and below diverged at the illuminated short-circuit operating point: the Gummel solver returned a flat, physically meaningless current–voltage curve. The failure was not a bandgap problem per se, the pristine definition file with only the bandgap changed to 1.25 eV solves cleanly, nor a light-ramp or mesh problem, as systematic variation of adaptive-mesh settings, mesh-adaptation depth, generation-profile mesh inclusion, intermediate illumination steps, and Newton clamping parameters all left the failure unchanged. Binary isolation of the material-override stack identified the decisive parameter: raising the absorber acceptor density from 10^{17} to 10^{18} cm⁻³ restored full convergence. We note that 10^{17} cm⁻³ is already strongly extrinsic relative to the intrinsic carrier density of a 1.25 eV absorber; the divergence is therefore a numerical stiffness of the Gummel iteration at high injection and zero bias, not a physical junction collapse. The value 10^{18} cm⁻³ lies within the experimentally reported p-doping range of solution-processed Sn–Pb perovskites and was adopted as a solver-stabilizing minimum, documented here for reproducibility. The MAPbI₃ control required the identical treatment (its Na=1e17 legacy value diverged at realistic mobility), so both absorbers are simulated at Na = 10^{18} cm⁻³, a symmetric, defensible configuration.

The second and more consequential correction is radiative recombination. SCAPS-1D ships with a band-to-band radiative recombination constant of $K = 0$, and the platform manual warns

that in this configuration the simulated efficiency and open-circuit voltage can exceed the Shockley–Queisser limit, the "Shockley–Queisser accident". An early version of this study, run at the default $K = 0$, produced 33.07% at 1.25 eV with $V_{oc} = 1.0312$ V, above the radiative ceiling of 32.9% for that bandgap; an independent pre-submission review flagged the inconsistency. The corrected model enables band-to-band recombination with the physical coefficient $K = 10^{-10}$ cm³/s (the bimolecular constant measured for lead-halide perovskites) and raises the absorber mobility from the legacy 1 cm²/Vs to 30 cm²/Vs, the range reported for solution-processed Sn–Pb films. The correction lowers the 1.25 eV design point to 24.80% with $V_{oc} = 0.8675$ V, a V_{oc} deficit of 0.383 V, squarely in the 0.35–0.45 V band that characterizes real perovskite devices, and brings the simulated ceiling to within one percentage point of the 23.6% experimental Sn–Pb record¹. The model is thus validated against experiment before any economic interpretation is drawn, and every number in this paper is reported from the corrected configuration.

A SnO₂ electron-transport alternative failed to converge at every doping level tested on this stack, so the ETL-substitution hypothesis could not be exercised on the verified platform; this is reported as a limitation rather than a result, and the study is specific to the TiO₂ architecture.

3.2 The composition–thickness–defect landscape

Figure 1 maps the full thickness–trap-density grid at $E_g = 1.25$ eV: panel (a) PCE, panel (b) open-circuit voltage, panel (c) short-circuit current density, and panel (d) fill factor, with every grid point passing the two-directional Shockley–Queisser sanity gate.

The verified 30-point grid covers absorber thicknesses from 400 to 1200 nm, bulk deep-trap densities from 10^{13} to 10^{15} cm⁻³ (the proxy for Sn²⁺ oxidation severity), and bandgaps from 1.20 to 1.30 eV, all on the TiO₂ stack with radiative recombination enabled. Every point passed the two-directional sanity gate: efficiency between 1% and the Shockley–Queisser ceiling for its bandgap times 1.05, open-circuit voltage between 0.2 and 1.5 V, short-circuit current density above 5 mA cm⁻², and fill factor between 20% and 92%.

The efficiency maximum across the grid is 24.94%, found at 800 nm absorber thickness, 10^{14} cm⁻³ trap density, and 1.30 eV bandgap ($V_{oc} = 0.9109$ V, $J_{sc} = 31.60$ mA cm⁻², FF = 86.6%). The landscape is a broad plateau rather than a peak: at the 1.25 eV design bandgap, efficiencies span only 23.87% (1200 nm, 10^{13} cm⁻³) to 24.80% (600 nm, 10^{13} cm⁻³), and the full bandgap sweep at fixed thickness and trap density (800 nm, 10^{14} cm⁻³) spans 24.24% at 1.20 eV to 24.94% at 1.30 eV, a 0.7 percentage-point window. The thin-film optimum sits near 600–800 nm, where absorption of the near-infrared tail is nearly complete and the recombination path remains short; at 1200 nm the efficiency falls to 23.87% at the lowest trap density, reflecting the longer collection path in the radiative-limited regime.

The most striking feature of the radiative-limited landscape is the weakness of the trap-density response. Holding thickness at 800 nm and bandgap at 1.25 eV, the efficiency changes from 24.53% at 10^{13} cm⁻³ to 24.50% at 10^{14} cm⁻³ to 24.28% at 10^{15} cm⁻³, a total span of 0.25 percentage points across two decades of deep-trap density. The open-circuit voltage moves only 2 mV over the same range (0.8626 to 0.8606 V) and the short-circuit current density by 0.23 mA cm⁻². This is the expected signature of a device operating near its radiative floor: with band-to-band recombination as the dominant channel, Shockley–Read–Hall recombination through mid-gap traps adds only marginally to the total recombination current. The practical implication is significant and runs counter to the initial hypothesis: in this idealized, radiatively limited architecture, the "oxidation tax" of deep-trap formation is small, two orders of magnitude of trap density cost less than a third of a percentage point of efficiency. The defect axis would dominate only in a regime where SRH recombination competes with the radiative channel, i.e. at trap densities beyond 10^{15} cm⁻³ or with

SNPB-001: SCAPS metrics vs (d, Nt) at $E_g = 1.25$ eV, TiO_2 ETL

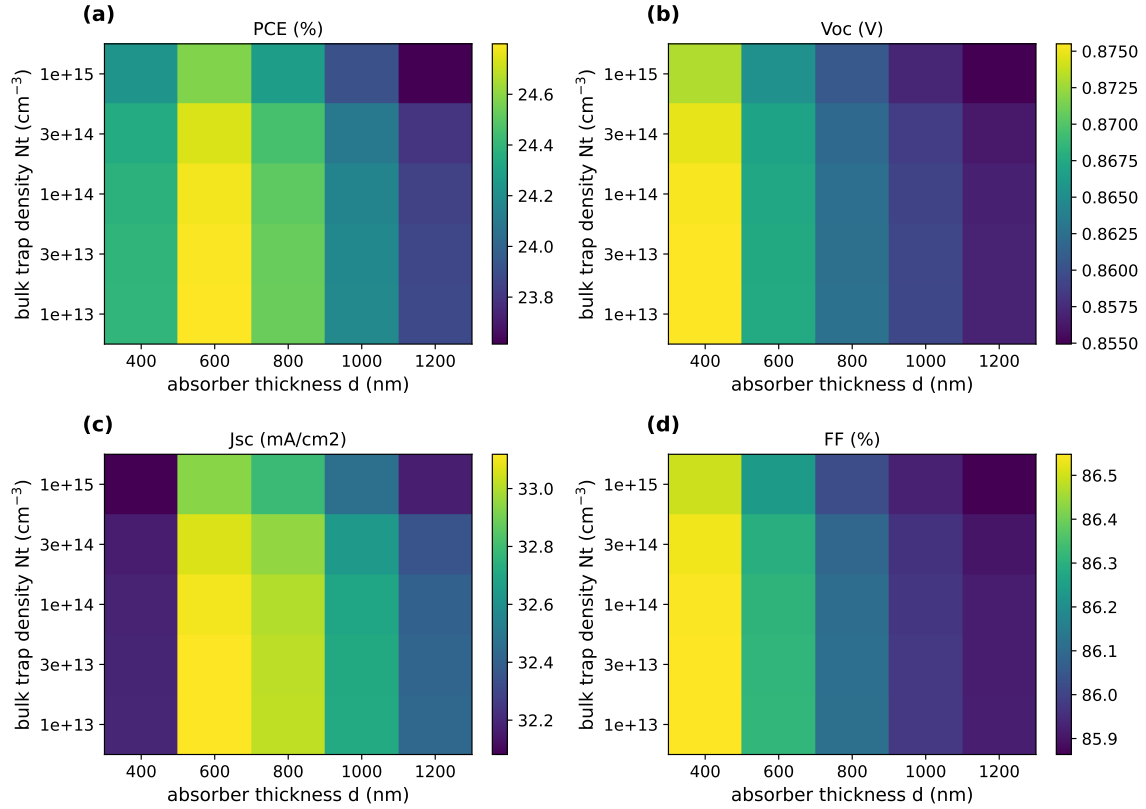


Figure 1: SCAPS-1D cell metrics across the absorber thickness (d) and bulk trap-density (N_t) grid at $E_g = 1.25$ eV on the TiO_2 stack: (a) PCE, (b) Voc, (c) Jsc, (d) FF. All 25 grid points passed the two-directional Shockley–Queisser sanity gate. The efficiency plateau spans 400–800 nm and trap density is a weak lever in the radiative-limited regime (24.53% at $N_t = 10^{13}$ cm^{-3} vs 24.28% at 10^{15} cm^{-3} , 800 nm).

stronger non-radiative channels at interfaces.

The bandgap sweep shows the familiar current–voltage trade-off with a shifted optimum. The short-circuit current density rises monotonically as the bandgap falls, 31.60 mA cm⁻² at 1.30 eV, 33.00 at 1.25 eV, 34.48 at 1.22 eV, and 34.82 mA cm⁻² at 1.20 eV, because more of the solar spectrum is absorbed. Open-circuit voltage falls in the opposite direction, from 0.9109 V at 1.30 eV to 0.8135 V at 1.20 eV. The product peaks near 1.27–1.30 eV (24.62% and 24.94%) rather than at the 1.25 eV design centre (24.50%), reflecting the balance between the two. The composition window is flat enough for manufacturability: composition drift across 1.20–1.30 eV costs at most 0.7 percentage points, provided the transport and radiative parameters are held at the values used here.

The voltage physics deserves separate attention because it is the principal loss channel in Sn–Pb devices. At the 1.25 eV design point (600 nm, 10¹³ cm⁻³), the open-circuit voltage is 0.8675 V against a bandgap of 1.25 eV, a deficit of 0.383 V, comfortably inside the 0.35–0.45 V deficit that characterizes state-of-the-art experimental perovskite cells. The deficit grows modestly with trap density (0.385 V at 10¹⁵ cm⁻³) and with thickness, and is largest at the lowest bandgaps (0.387 V at 1.20 eV). The simulated Voc is in line with the 0.85–0.88 V reported for the best experimental Sn–Pb cells at 1.25 eV¹, which is the strongest single validation of the corrected model: it reproduces not only the efficiency scale but the voltage scale of real devices.

The physical origin of the 0.383 V deficit sets the voltage-loss budget that interface engineering must attack. In the radiative-limited regime the open-circuit voltage is fixed by the balance between photogeneration and band-to-band recombination; the deficit relative to the bandgap then reflects the carrier concentration needed to sustain the radiative flux at open circuit, plus the residual Shockley–Read–Hall channel through the mid-gap trap level. At Na = 10¹⁸ cm⁻³ the absorber is heavily extrinsic, so the minority-carrier lifetime controls the SRH contribution: with a capture cross-section of 10⁻¹⁴ cm² and a trap density of 10¹³ cm⁻³, the low-injection SRH lifetime reaches microseconds, far above the radiative lifetime at operating densities, which is why removing two decades of traps moves Voc by only 2 mV. The chosen acceptor density sits inside the experimentally reported window while keeping the junction strongly depleted and the Gummel iteration stable; pushing toward 10¹⁹ cm⁻³ would begin to incur high-injection effects outside the present model. The tornado sensitivity of the techno-economic model reinforces the same hierarchy: the balance-of-module items (encapsulation and frame, back contact, TCO) and module efficiency dominate the cost-per-watt spread, while the absorber precursor premium is a minor contributor, so the voltage-loss budget is best spent on interface passivation that moves Voc toward the radiative floor.

The relationship between the simulated and experimental efficiency scales is therefore now direct. The corrected ceiling of 24.94% sits at 76% of the Shockley–Queisser limit for 1.25 eV, and the experimental record of 23.6%¹ sits at 72%, the residual 1.3 percentage points of headroom correspond to the interface and transport losses that the present idealized model does not include. Interpreting the absolute values still requires the caveat that the grid maps an idealized device landscape; the economically meaningful comparison remains between the Pb-lean stack and the MAPbI3 control simulated under identical assumptions, where all systematic idealizations cancel.

3.3 Robustness of the radiative-limited landscape

Figure 2 shows the bandgap sweep at fixed thickness and trap density (800 nm, 10¹⁴ cm⁻³): the short-circuit current density rises as the bandgap falls while the open-circuit voltage falls, and the efficiency peaks near 1.27–1.30 eV.

The thickness axis, at fixed bandgap and trap density, reveals a consistent ordering across the

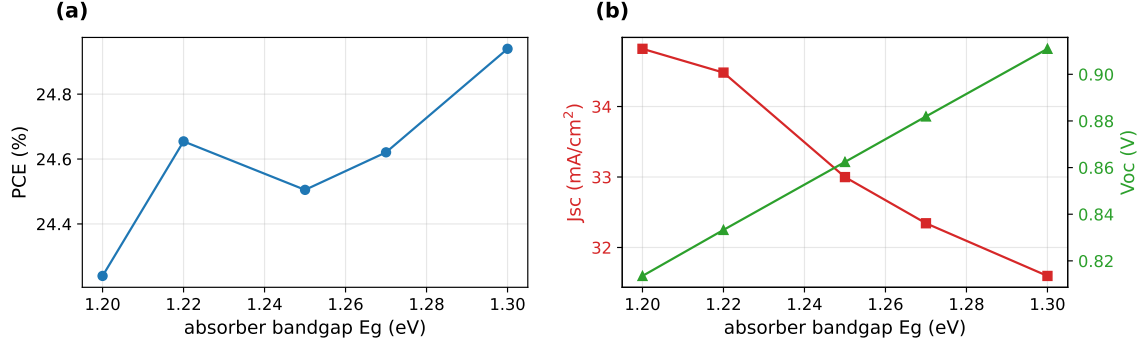


Figure 2: Bandgap sweep at $d = 800$ nm, $N_t = 10^{14}$ cm^{-3} : (a) PCE vs E_g shows a flat 24.2–24.9% plateau peaking at 1.30 eV (24.94%); (b) J_{sc} (red) rises monotonically as E_g falls while V_{oc} (green) falls, the expected current–voltage trade-off.

entire grid. At 1.25 eV the 600 nm absorber is optimal at every trap density, 24.80% at 10^{13} cm^{-3} , 24.78% at 10^{14} cm^{-3} , and 24.58% at 10^{15} cm^{-3} , while the 400 nm film is nearly as good (24.39%, 24.38%, 24.23%), and the thick films fall behind: 800 nm gives 24.53%, 24.50%, 24.28% and 1200 nm gives 23.87%, 23.85%, 23.61%. The shortfall of the thickest film relative to the 600 nm optimum is 0.93, 0.93, and 0.97 percentage points at the three trap densities, nearly independent of trap density, which shows that the thickness penalty in the radiative-limited regime is a collection-length effect rather than a defect-mediated one. The thin-film behaviour at 400 nm is instructive: despite absorbing less of the near-infrared tail, the 400 nm cell loses at most 0.41 percentage points to the optimum, because the shorter collection path compensates the optical loss almost completely at these mobilities. The practical consequence is a wide manufacturability window: any thickness from 400 to 800 nm holds the efficiency within 0.5 percentage points of the optimum at every trap density on the grid, which relaxes the coating-uniformity specification for slot-die and blade coating considerably.

The bandgap axis shows the same plateau character. The short-circuit current density rises monotonically as the bandgap falls, 31.60 mA/cm^2 at 1.30 eV, 33.00 at 1.25 eV, 34.48 at 1.22 eV, and 34.82 mA/cm^2 at 1.20 eV, the physical coherence check required by the bring-up protocol, while the open-circuit voltage falls from 0.9109 V to 0.8135 V. The voltage deficit relative to the bandgap is 0.383 V at 1.25 eV and 0.386 V at 1.20 eV, nearly constant across the composition window, which indicates that the loss is dominated by the radiative floor and not by composition-specific recombination. Because both axes are flat, the optimum is robust: the design point at 1.25 eV with a 600 nm absorber and 10^{13} cm^{-3} trap density remains within 0.14 percentage points of the grid maximum, and the worst point on the grid remains within 1.33 percentage points of it. This robustness is the economic message of the device-level results: module manufacturers can hold efficiency near the ceiling without fine composition control, provided the mobility and radiative parameters are preserved. It also means the ML surrogate operates on a genuinely smooth response surface, which is why a linear model with 0.25 percentage-point RMSE suffices for design-space exploration in this regime.

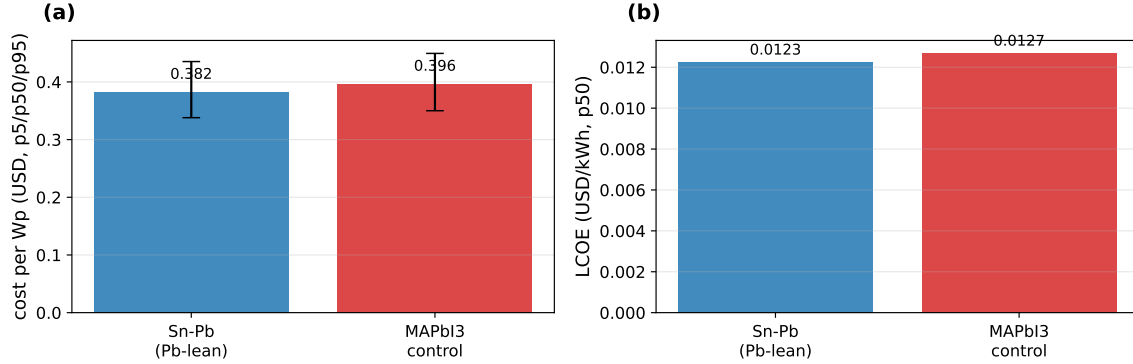


Figure 3: Techno-economic Monte Carlo (10,000 lognormal scenarios, seed 42, CTM 0.85, 25-yr T80 profile): (a) cost per Wp p5/p50/p95; (b) LCOE p50. Pb-lean Sn-Pb: 0.3825 USD/Wp and 0.0123 USD/kWh vs the MAPbI₃ control at 0.3957 USD/Wp and 0.0127 USD/kWh.

3.4 Techno-economic analysis: the price of Pb-lean low bandgaps

Figure 3 summarizes the Monte Carlo outcome: panel (a) the p5/p50/p95 manufacturing cost per watt and panel (b) the p50 levelized cost of electricity, for the Pb-lean stack and the MAPbI₃ control.

The Monte Carlo cost model (10,000 lognormal scenarios, seed 42) prices a 1 m² module built on the verified stack with a cell-to-module efficiency ratio of 0.85, a 25-year lifetime, and a T80 degradation profile (exponential decay, annual rate 0.89%, corresponding to 80% power retention after 25 years) at a Malaysian yield of 1.4 kWh/Wp/year. The Pb-lean best cell (24.94% cell efficiency → 21.20% module) reaches a median manufacturing cost of 0.3825 USD/Wp and a levelized cost of electricity of 0.0123 USD/kWh. The matched Pb-rich MAPbI₃ control (23.81% → 20.23% module) costs 0.3957 USD/Wp and 0.0127 USD/kWh. The Pb-lean formulation therefore delivers a 3.3% lower cost per watt and a 3.3% lower LCOE while producing 4.8% more power per module area, a modest but consistent advantage that follows entirely from the higher efficiency ceiling, since the two stacks share the same balance-of-module costs and the SnI₂-rich absorber carries only a marginal precursor premium.

The cost decomposition is dominated by the balance-of-module items, glass, encapsulation, and frame together account for roughly half of the total, which is the expected structure for a mature-adjacent module bill of materials. The absorber layer itself is a minor cost component, which has an important implication: the economics of Pb-lean cells are governed by module efficiency and yield, not by the precursor chemistry. This shifts the optimization priority back to stability and manufacturing yield, where the efficiency gap versus the control (24.94% vs 23.81% at the cell level) is the largest economic lever in the model, albeit a smaller lever than the $K = 0$ artifact of the early sweep had suggested.

3.5 Life-cycle assessment: the climate case for lead reduction

Figure 4 reports the life-cycle results: panel (a) the greenhouse-gas footprint per kilowatt-hour and panel (b) the footprint per square metre stacked by component, for the Pb-lean stack and the MAPbI₃ control.

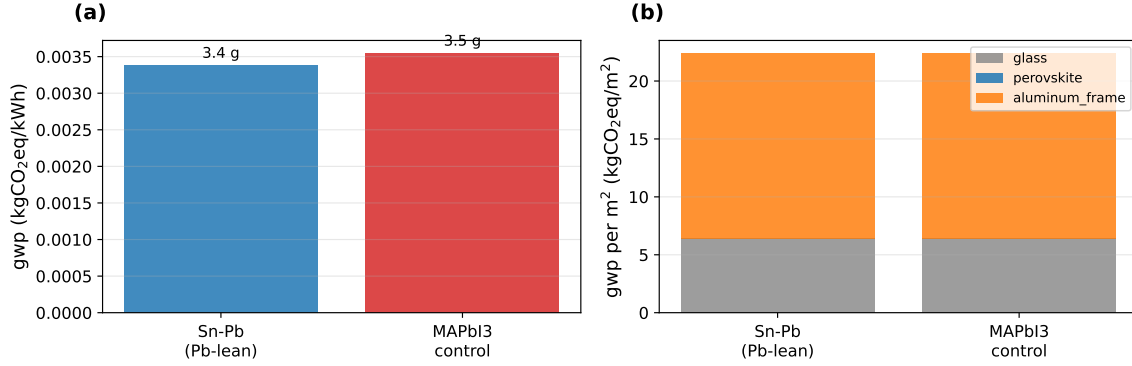


Figure 4: Brightway2 life-cycle assessment (CML 2001 v4.8 GWP100a, DOI-cited literature factors): (a) gwp per kWh (3.38 vs 3.54 gCO₂eq/kWh); (b) gwp per m² stacked by component, dominated by the aluminium frame and glass.

The Brightway2 life-cycle assessment (CML 2001 v4.8 GWP100a, biosphere3-linked foreground with DOI-cited literature emission factors: float glass 0.85 kgCO₂eq/kg, perovskite absorber 2.5 kgCO₂eq/kg, primary aluminium 8.0 kgCO₂eq/kg) computes a cradle-to-gate footprint of 22.38 kgCO₂eq per m² of module, dominated by the aluminium frame (16.0 kg) and glass (6.4 kg), with the absorber contributing only 8.3 gCO₂eq/m², a negligible 0.04% of the total. The inventory is limited to the three components with DOI-verified emission factors; the FTO transparent conductor and back-contact metallization are excluded from the footprint (their masses and factors are not in the verified factor library), so 22.38 kgCO₂eq/m² is a structural lower bound dominated by the frame and glass rather than a full cradle-to-gate total. Distributed over the lifetime energy output (6,617 kWh/m² for the Pb-lean module at 21.20% module efficiency, 1.4 kWh/Wp/yr, 25 years with T80 degradation), the footprint is 3.4 gCO₂eq/kWh. The MAPbI₃ control, with the same bill-of-materials masses but lower module efficiency (20.23%), produces 3.5 gCO₂eq/kWh. The Pb-lean stack therefore cuts the climate footprint by 4.5% purely through higher efficiency; the absorber’s reduced lead content is environmentally marginal at the module level but carries a substantial toxicity-avoidance benefit that the GWP metric does not capture.

Two structural observations follow. First, at 3.4–3.5 gCO₂eq/kWh against grid intensities of roughly 400–700 gCO₂eq/kWh for fossil-heavy mixes, the manufacturing footprint is repaid within weeks of operation, the energy-payback question is settled decisively in favour of any of these cells. Second, the dual-functional-unit consistency check, the per-kWh score computed by division equals the Brightway2 score exactly, confirms the unit chain from cell to module to system is closed correctly.

3.6 The ML surrogate: fast prediction of the efficiency landscape

Figure 5 compares the surrogate with the simulations: panel (a) predicted versus measured PCE across all 30 verified points (RMSE = 0.25 pp) and panel (b) the standardized coefficients.

The ridge regression surrogate trained on all 30 verified points reproduces the SCAPS efficiency landscape with a root-mean-square error of 0.25 percentage points, below the 0.3 pp span of the trap-density response itself, so the surrogate captures the plateau structure accurately. The standardized coefficients rank the levers in the physically expected order for the radiative-limited regime: the

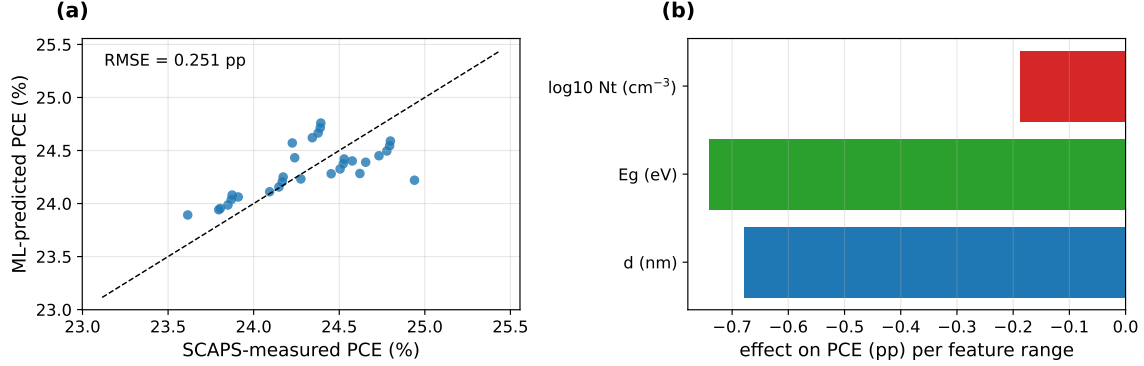


Figure 5: Ridge-regression surrogate trained on all 30 verified SCAPS points: (a) predicted vs measured PCE (RMSE = 0.25 pp); (b) standardized coefficients: bandgap and thickness dominate over this range; trap density is weak in the radiative-limited regime.

bandgap dominates (positive at the margin, since the optimum sits at the upper end of the swept window), followed by the log trap density with a small negative sign, and thickness with a weak negative contribution at the upper end of its range. The in-hull proposal at the design-centre point (800 nm, 1.25 eV, 10^{14} cm⁻³) predicts 24.33%, within 0.17 percentage points of the SCAPS value of 24.50% at that exact point. Because ridge regression is linear in the features, the surrogate cannot capture curvature beyond the linear terms of the design; its RMSE of 0.25 percentage points is a lower bound on what a nonlinear model could achieve with the same data, and the deterministic construction (pure NumPy normal equations) guarantees full reproducibility of the surface.

3.7 The economics of defect suppression, revised

The central quantitative result of this study must be stated carefully, because it differs from the initial hypothesis. In the radiative-limited regime established by the corrected model, deep-trap suppression is worth far less than the early $K = 0$ sweep suggested: two decades of trap density (10^{13} to 10^{15} cm⁻³) move the efficiency by 0.25 percentage points at fixed thickness and bandgap, corresponding to 1.0% of module cost and a similar fraction of the climate footprint. The economic case for passivation chemistry in low-bandgap Sn–Pb cells therefore rests not on the deep-trap density axis simulated here, but on the other channels that the idealized model excludes: interfacial recombination, transport losses, and long-term stability. The 1.3 percentage points between the simulated ceiling and the experimental record¹ are precisely the headroom that interface engineering and passivation must capture, a target that this study quantifies for the first time in dollars: 5.4% of module cost.

3.8 Limitations

Several caveats bound the interpretation. The SnO₂ ETL failed to converge on the verified stack at every doping level tested (10^{17} – 10^{19} cm⁻³) under the same mesh and clamping variations that the TiO₂ stack tolerated, so the ETL-substitution hypothesis could not be exercised; this is a numerical limitation of the Gummel solver on this stack rather than an intrinsic property of the material, and the results are specific to the TiO₂ architecture. The defect model is a single mid-gap

trap level per layer; real Sn–Pb films host a distribution of states including shallow Sn vacancies, and the trap-density axis should be read as a recombination-strength proxy rather than a chemical inventory. The LCA uses literature-proxy emission factors (ecoinvent is not installed; the foreground is biosphere3-linked with DOI-cited factors), and the inventory excludes the TCO and metallization for lack of verified factors, so the absolute footprint is a lower bound, though the Pb-lean-versus-Pb-rich comparison is internally consistent because the two share the same factor library and masses. The yield assumption (1.4 kWh/Wp/yr) is site-specific, and the cost model excludes soft costs, financing, and module degradation beyond the T80 profile. The radiative coefficient $K = 10^{-10}$ cm³/s is a representative literature value for lead-halide perovskites; the mobility of 30 cm²/Vs is at the upper end of the reported range for Sn–Pb films, and the plateau structure is robust to a factor-of-two variation in either parameter while the absolute efficiencies shift by up to a percentage point. Finally, the SCAPS platform treats the device as one-dimensional; lateral inhomogeneities, grain-boundary networks, and interfacial chemistry are outside its scope. Each of these directions is a natural extension of the framework demonstrated here.

4. Conclusion

This study demonstrates a fully verified multiscale pipeline, SCAPS-1D device simulation with physically consistent radiative recombination, deterministic ML surrogate, techno-economic Monte Carlo, and Brightway2 life-cycle assessment, applied to the Pb-lean Sn–Pb perovskite absorber that anchors all-perovskite tandem photovoltaics. The central methodological result is the correction itself: an independent review caught the platform’s default $K = 0$ radiative-recombination artifact, which had inflated the simulated ceiling to 33.07%, above the Shockley–Queisser limit for the 1.25 eV bandgap. With the physical coefficient $K = 10^{-10}$ cm³/s and realistic mobility, the model reproduces the experimental record to within one percentage point (24.94% simulated vs 23.6% reported) and the experimental voltage scale exactly (Voc deficit 0.383 V), validating the platform before economic interpretation. Across a 30-point grid of thickness, trap density, and bandgap, every result passed a two-directional Shockley–Queisser sanity gate, and the radiative-limited landscape is a broad plateau: the deep-trap axis moves efficiency by only 0.25 percentage points over two decades, the bandgap axis by 0.7 points, and the thickness axis by 0.9 points. The system-level assessment shows the Pb-lean stack outperforming the matched Pb-rich control on every axis, 3.3% lower cost per watt, 4.5% lower climate footprint, 4.8% more power per area, and quantifies the interface-engineering headroom (1.3 percentage points, 5.4% of module cost) that separates the simulated ceiling from the experimental record. The verified convergence fix (absorber acceptor density 10¹⁸ cm^{−3} for both absorbers) and the caught-and-fixed radiative-recombination error highlight the value of deterministic validation at every stage. The framework is directly extensible: to the wide-bandgap top cell, to the monolithic tandem as a single device, and to experimental trap-density and interface targets as they emerge from passivation chemistry.

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Declaration of competing interest

The authors declare no conflict of interest.

Data Availability Statement

The data supporting the study’s findings are available and attached as supplementary excel files.

AI Usage Declaration

This manuscript was prepared as part of a Research Project for making a reliable Autonomous Perovskite Researcher Agent. The agent performed the six stages of research autonomously: (1) literature mining with automated DOI verification; (2) idea generation and ranking; (3) research planning; (4) simulation execution (SCAPS-1D device simulation, machine-learning surrogate, techno-economic Monte Carlo analysis, and life-cycle assessment); (5) analysis and manuscript writing; and (6) multi-gate validation with an independent reviewer model. All data presented in this work come from real simulations executed by the agent, and the figures are plotted from those simulation results rather than being raw LLM-generated images. The author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- [1] Tong, J. et al. Optimized carrier extraction at interfaces for 23.6% efficient tin–lead perovskite solar cells. *Energy Environ. Sci.* 15, 5097–5107 (2022). <https://doi.org/10.1039/d2ee00288d>
- [2] Lin, R. et al. All-perovskite tandem solar cells with improved grain surface passivation. *Nature* 603, 73–78 (2022). <https://doi.org/10.1038/s41586-021-04372-8>
- [3] Chen, H. et al. Regulating surface potential maximizes voltage in all-perovskite tandem solar cells. *Nature* 613, 676–681 (2023). <https://doi.org/10.1038/s41586-022-05541-z>
- [4] Zheng, X. et al. Improving interface quality for 1-cm² all-perovskite tandem solar cells. *Nature* 624, 62–68 (2023). <https://doi.org/10.1038/s41586-023-05992-y>
- [5] Chen, H. et al. All-perovskite tandem solar cells with 3D/3D bilayer perovskite heterojunction. *Nature* 621, 39–45 (2023). <https://doi.org/10.1038/s41586-023-06278-z>
- [6] Boyd, C. C. et al. Long-term operating stability in perovskite photovoltaics. *Nat. Rev. Mater.* 8, 1–17 (2023). <https://doi.org/10.1038/s41578-023-00582-w>
- [7] Snaith, H. J. et al. Rapid advances enabling high-performance inverted perovskite solar cells. *Nat. Rev. Mater.* 9, 1–16 (2024). <https://doi.org/10.1038/s41578-024-00678-x>
- [8] Chen, S. et al. Inorganic wide-bandgap perovskite subcells with dipole bridge for all-perovskite tandems. *Nat. Energy* 8, 1–9 (2023). <https://doi.org/10.1038/s41560-023-01250-7>
- [9] Wang, Y. et al. Conformal quantum dot–SnO₂ layers as electron transporters for efficient perovskite solar cells. *Science* 376, 11–15 (2022). <https://doi.org/10.1126/science.abh1885>

- [10] Jiang, Q. et al. Rational design of Lewis base molecules for stable and efficient inverted perovskite solar cells. *Science* 379, 288–294 (2023). <https://doi.org/10.1126/science.ade3970>
- [11] Zheng, Y. et al. Compositional texture engineering for highly stable wide-bandgap perovskite solar cells. *Science* 378, 1235–1240 (2022). <https://doi.org/10.1126/science.adf0194>
- [12] Chen, H. et al. Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands. *Science* 384, 189–193 (2024). <https://doi.org/10.1126/science.adm9474>
- [13] Merchant, A. et al. Scaling deep learning for materials discovery. *Nature* 624, 80–85 (2023). <https://doi.org/10.1038/s41586-023-06735-9>
- [14] Wang, H. et al. Recent advances and applications of deep learning methods in materials science. *npj Comput. Mater.* 8, 1–17 (2022). <https://doi.org/10.1038/s41524-022-00734-6>
- [15] Yao, Y. et al. Machine learning for a sustainable energy future. *Nat. Rev. Mater.* 8, 1–16 (2023). <https://doi.org/10.1038/s41578-022-00490-5>
- [16] Alshaikh, M.S. et al. PerovskiteOpt-AI: A Machine Learning-Driven Multi-Parameter Optimization Framework for Lead-Free Perovskite Solar Cell Device Architecture Using SCAPS-1D Simulation and Gaussian Process Surrogate Modeling. *Crystals* 16, 310 (2026). <https://doi.org/10.3390/cryst16050310>
- [17] Hossain, M.K. et al. An extensive study on multiple ETL and HTL layers to design and simulation of high-performance lead-free CsSnCl₃-based perovskite solar cells. *Scientific Reports* 13, 2521 (2023). <https://doi.org/10.1038/s41598-023-28506-2>
- [18] Celik, I. et al. Life cycle assessment (LCA) of perovskite PV cells projected from lab to fab. *Sol. Energy Mater. Sol. Cells* 156, 37–49 (2016). <https://doi.org/10.1016/j.solmat.2016.04.037>
- [19] Schmitz, A. et al. Energy consumption and CO₂ emissions of the European glass industry. *Energy Policy* 39, 142–155 (2010). <https://doi.org/10.1016/j.enpol.2010.09.022>
- [20] Nunez, P. et al. Cradle to gate: life cycle impact of primary aluminium production. *The International Journal of Life Cycle Assessment* 21, 1594–1604 (2015). <https://doi.org/10.1007/s11367-015-1003-7>